

Mohammadreza Teymoorianfard

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Education

University of Massachusetts Amherst

PHD IN COMPUTER SCIENCE

- GPA: 3.88/4

Massachusetts, USA

Sep 2023 - present

MSC IN COMPUTER SCIENCE

- GPA: 3.87/4

Sep 2023 - May 2025

University of Tehran

BSc IN ELECTRICAL ENGINEERING

- GPA: 19.06/20 (4/4)

Tehran, Iran

Sep 2018 - Jun 2023

MINOR IN COMPUTER ENGINEERING

Sep 2019 - Jun 2023

Research Interests

- Trustworthy Machine Learning
- Privacy and Security of Embodied AI Systems
- Privacy and Security of Generative AI Models
- Privacy and Security of AI Agents

Publications

Teymoorianfard, M., Monteuuis, J. P., Petit, J., and Houmansadr, A. (2026). *ReasonBreak: Probing Vulnerabilities in Reasoning-Enabled Vision-Language-Action Models for Autonomous Driving*. arXiv preprint arXiv:2605.29114.

Mittapalli, V., Dani, S. J., Pili, S. S., Ansu, S., **Teymoorianfard, M.**, Dernoncourt, F., ... and Ahmed, N. K. (2026). *TRACE: Trajectory Reasoning through Adaptive Cross-Step Evidence Aggregation for LLM Agents..* arXiv preprint arXiv:2606.07054.

Jeong, H., **Teymoorianfard, M.**, Kumar, A., Houmansadr, A. and Badasarian, E., 2025. *Network-Level Prompt and Trait Leakage in Local Research Agents*. arXiv preprint arXiv:2508.20282. (Accepted to USENIX Security '26)

Teymoorianfard, M., Ma, S. and Houmansadr, A., 2025. *VIDSTAMP: A Temporally-Aware Watermark for Ownership and Integrity in Video Diffusion Models*. arXiv preprint arXiv:2505.01406.

Amini, S., **Teymoorianfard, M.**, Ma, S. and Houmansadr, A., 2024. *MeanSparse: Post-Training Robustness Enhancement Through Mean-Centered Feature Sparsification*. arXiv preprint arXiv:2406.05927.

Research Experience

Umass Amherst - The Secure, Private Internet (SPIN) Research Group

Amherst, MA

ADVISOR: AMIR HOUMANSADR

Sep. 2023 - Present

- **Reasoning Vulnerabilities in VLA Models** — Developed ReasonBreak, a benchmark and evaluation framework for analyzing reasoning-safety vulnerabilities in autonomous-driving Vision-Language-Action models.
- **Privacy Analysis of Local Web/Research Agents** — Exposed vulnerabilities in locally deployed agents enabling prompt and trait recovery from observed IP activity, and proposed a coherent synthetic-query defense.
- **Watermarking and Attribution for Video Generation Models** — Created watermarking and model attribution techniques for video generation systems, enhancing security and traceability of generative models.
- **Robustness in Neural Networks** — Increased neural network robustness by reducing feature variation, achieving state-of-the-art AutoAttack accuracy on CIFAR-10, CIFAR-100, and ImageNet.

University of Tehran - Smart Networks Lab

Tehran, Iran

ADVISOR: HAMED KEBRAIEI

Jun. 2022 - Jun. 2023

- Implemented a Model Predictive Control (MPC) for autonomous taxi navigation.

University of Tehran

ADVISOR: RESHAD HOSSEINI

- Developed a text detection system for card images, enhancing accuracy in document recognition.

Tehran, Iran

Jun. 2021 - Sep. 2021

Skills

Programming Languages

Python, C/C++, MATLAB, Verilog

Python Libraries & Frameworks

PyTorch, TensorFlow, Transformers, OpenCV, scikit-learn, NumPy, Pandas, Matplotlib, RL-Glue, PuLP, MIP

Honors & Awards

- 2023 **Ranked 3rd among 120 Electrical Engineering students**
- 2023 **Recipient of an industry-sponsored award for outstanding bachelor's thesis work**
- 2020 **Awarded the University of Tehran Sponsors Foundation Honorable Award for Academic Excellence**
- 2020 **Recipient of the Faculty of Engineering (FOE) Award for achieving 2nd rank in the 2019-2020 academic year**
- 2018 **Ranked in the top 0.4% of over 150,000 students in the Iranian National University Entrance Exam**
- 2011 **Admitted to National Organization for Exceptional Talents (NODET) for middle and high school**

Teaching Experience

Fall '24 & Spr '25 & '26	CICS 160: Object-Oriented Programming, TA	<i>Umass Amherst</i>
Fall '25	CICS 110: Foundations of Programming with Python, TA	<i>Umass Amherst</i>
Summer '24	COMPSCI 589: Machine Learning, TA	<i>Umass Amherst</i>
Spring '24	COMPSCI 119: Intro to Programming, TA	<i>Umass Amherst</i>
Fall '22	Intelligent Systems, TA	<i>University of Tehran</i>
Spring '22	Mechatronics, TA	<i>University of Tehran</i>
Spring '22	Signal and Systems TA	<i>University of Tehran</i>
Fall '21	Engineering Probability and Statistics, TA	<i>University of Tehran</i>

Relevant Courses

University of Massachusetts Amherst

- COMPSCI646: Information Retrieval
- COMPSCI603: Robotics
- COMPSCI611: Advanced Algorithms
- COMPSCI685: Adv Natural Language Processing
- COMPSCI682: Neural Networks, Modern Intro
- COMPSCI660: Advanced Information Assurance

University of Tehran

- Deep Learning
- Reinforcement Learning
- Machine Learning
- Artificial Intelligence
- Mechatronics
- Linear Algebra
- Engineering Probability and Statistics
- Modern Control Systems

Language

- ◇ ENGLISH: Advanced Proficiency
- ◇ PERSIAN: Native